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(54) Title: PROCESS FOR THE PREPARATION OF PONESIMOD AND ITS INTERMEDIATES THEREOF

(57) Abstract: The present invention relates to a process for the preparation of Ponesimod of formula I and its intermediates thereof.

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**“PROCESS FOR THE PREPARATION OF PONESIMOD AND ITS
INTERMEDIATES THEREOF”**

Related applications:

This application claims the benefit under Indian Provisional Application Nos. 202241007153, filed on Feb 10, 2022 and 202241041742, filed on Jul 21, 2022; the contents of which are incorporated by reference herein.

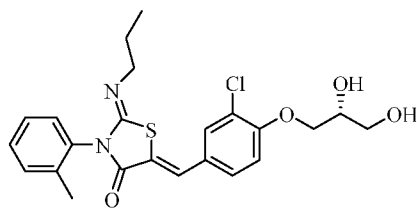
Field of the invention:

The present invention relates to a process for the preparation of Ponesimod of formula I and its intermediates thereof.

Background of the invention:

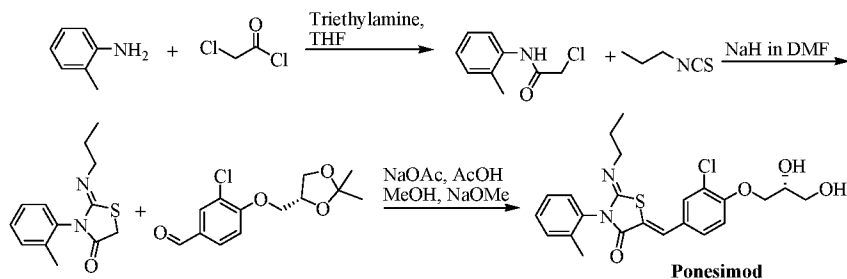
Ponesimod is a sphingosine 1-phosphate receptor modulator. It was approved and marketed as Ponvory in the United States and Europe in 2021 for the treatment of relapsing forms of multiple sclerosis, by Janssen pharmaceuticals.

Ponesimod is chemically known as (2Z,5Z)-5-[3-chloro-4-[(2R)-2,3-dihydroxypropoxy]benzylidene]-3-(2-methylphenyl)-2-(propylimino)-1,3-thiazolidin-4-one and is represented by the following structural formula I.



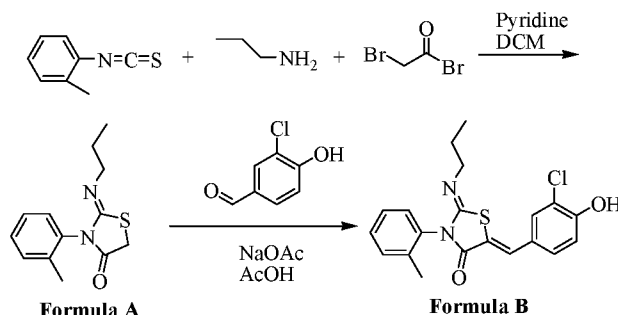
Formula I

Ponesimod and process for its preparation was first disclosed in the US7435828 (the US '828 patent) and the disclosed process is schematically depicted in the following scheme:

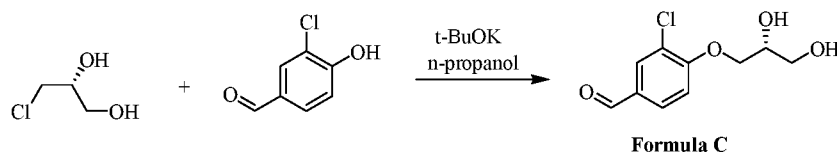


The US 8912340 patent (the US '340 patent) discloses a generic process for the preparation of Ponesimod intermediate compound of Formula A, chemically known as 3-(2-methyl phenyl)-2-((Z)-propylimino)thiazolidin-4-one and Formula B, chemically known as 5-(3-Chloro-4-hydroxy-benz-(Z)-ylidene)-2-[(Z)-propylimino]-3-(o-tolyl)-thiazolidin-4-one.

5 The disclosed process is schematically depicted in the following scheme:



The US 9340518 patent (the US '518 patent) discloses a process for the preparation of Ponesimod intermediate compound of Formula C, chemically known as (R)-3-chloro-4-(2,3-dihydroxypropoxy)-benzaldehyde. The disclosed process is schematically depicted in the following scheme:



The reported processes for the preparation of Ponesimod and its intermediates suffer from many disadvantages which include difficulty in achieving desired chemical and enantiomeric purity, tedious & cumbersome work up procedures, column chromatographic separations and purifications etc. All these disadvantages affect the overall yield as well as the quality of the final product.

Still there is a need for an alternative process for the preparation of Ponesimod which is economical, easy to scale up and commercially viable. Thus, main objective of the present invention is to provide an alternative process for the preparation of Ponesimod which avoids the shortcomings of the prior art.

Summary of the invention:

Accordingly, the present invention provides a process for the preparation of Ponesimod of formula I via novel intermediates.

5 In accordance with one embodiment, the present invention provides a process for the preparation of Ponesimod of formula I or its pharmaceutically acceptable salts, which comprises:

- a) reacting a compound of formula XII with a compound of general formula XIII in the presence of a base in a suitable solvent to provide a compound of general formula IV,
- 10 b) condensing the compound of general formula IV with a compound of formula VI in the presence of a base in a suitable solvent to provide a compound of general formula II, and
- c) converting the compound of general formula II into Ponesimod of formula I or its pharmaceutically acceptable salts.

15 In accordance with another embodiment, the present invention provides a process for the preparation of compound of formula IVa, which comprises,

- a) reacting 1,2-diol compound of formula XVIII with 2,2-dimethoxypropane in the presence of a suitable copper catalyst and a solvent to provide a compound of formula
- 20 XIIIb, and
- b) condensing the compound of formula XIIIb with a compound of formula XII in the presence of a suitable base and a solvent to provide a compound of formula IVa.

In accordance with another embodiment, the present invention provides a process for the preparation of compound of formula VI, which comprises

- a) reacting a compound of formula XIX with carbon disulfide in the presence of a suitable base to provide an *o*-tolylisothiocyanate of formula XX,
- b) reacting the *o*-tolylisothiocyanate of formula XX with *n*-propylamine in the presence of a suitable solvent to provide a compound of formula XXI; and
- 30 c) cyclizing the compound of formula XXI obtained in step-(b) with 2-chloro-acetyl chloride in the presence of a suitable base and a solvent to provide a compound of formula VI.

In accordance with another embodiment, the present invention provides an alternate process for the preparation of compound of formula VI, which comprises

- a) reacting n-propylamine with carbon disulfide in the presence of a suitable base to provide a n-propyl isothiocyanate of formula XXII,
- 5 b) reacting the n-propyl isothiocyanate of formula XXII with a compound of formula XIX in the presence of a suitable solvent to provide a compound of formula XXI, and
- c) cyclizing the compound of formula XXI obtained in step-(b) with 2-chloro-acetyl chloride or 2-bromo-acetyl bromide in the presence of a suitable base and a solvent to provide a compound formula VI.

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In accordance with another embodiment, the present invention provides a process for the preparation of Ponesimod of formula I, which comprises,

- 1) preparing a compound of formula IVa, or a compound of formula VI according the present invention process; and
- 15 2) converting the compound of formula IVa or a compound of formula VI into Ponesimod of formula I.

In accordance with another embodiment, the present invention provides a process for the preparation of Ponesimod of formula I or its pharmaceutically acceptable salts, which
20 comprises:

- a) condensing a compound of formula V with a compound of general formula IV in the presence of a base in a suitable solvent to provide a compound of general formula III,
- b) reacting the compound of general formula III with n-propylhalide to provide a compound of general formula II, and
- 25 c) converting the compound of general formula II into ponesimod of formula I or its pharmaceutically acceptable salts.

In accordance with another embodiment, the present invention provides a process for the preparation of Ponesimod of formula I or its pharmaceutically acceptable salts, which
30 comprises:

- a) reacting compound of formula XII with alkyl or aryl sulfonyl halide in the presence of a base in a suitable solvent to provide a compound of general formula XIV,

- b) condensing the compound of general formula XIV with a compound of formula VI to provide a compound of general formula VII, and
- c) converting the compound of general formula VII into ponesimod of formula I or its pharmaceutically acceptable salts.

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In accordance with another embodiment, the present invention provides a process for the preparation of Ponesimod of formula I or its pharmaceutically acceptable salts, which comprises:

- a) reacting a compound of formula V with a compound of formula XII to provide a
10 compound of formula IX,
- b) reacting the compound of formula IX with alkyl or aryl sulfonyl halide in the presence of a base in a suitable solvent to provide a compound of general formula VIII,
- c) reacting the compound of general formula VIII with n-propylhalide to provide a compound of general formula VII, and
- 15 d) converting the compound of general formula VII into ponesimod of formula I or its pharmaceutically acceptable salts.

In accordance with another embodiment, the present invention provides a process for the preparation of Ponesimod of formula I or its pharmaceutically acceptable salts, which
20 comprises:

- a) reacting a compound of formula XI with a compound of general formula XIII in the presence of a base in a suitable solvent to provide a compound of general formula X,
- b) converting the compound of general formula X into the compound of general formula II by reacting with a suitable chlorinating agent; and
- 25 c) converting the compound of general formula II into ponesimod of formula I or its pharmaceutically acceptable salts.

In accordance with another embodiment, the present invention provides a process for the preparation of Ponesimod of formula I or its pharmaceutically acceptable salts, which
30 comprises:

- a) condensing the compound of formula XV with a compound of formula V or a compound of formula VI to provide a corresponding compound of formula XVI or compound of formula XVII,
- b) converting the compound of formula XVI and compound of formula XVII into the compound of formula IIA, and
- c) converting the compound of formula IIA into ponesimod of formula I or its pharmaceutically acceptable salts.

In accordance with another embodiment, the present invention provides novel intermediate compound of formula IIA, III, IVA, VII, VIII, IX, X, XVI and XVII.

In accordance with another embodiment, the present inventions provides use of novel compounds of formula IIA, III, IVA, VIIA, VIIB, VIII, IX, X, XVI and XVII as an intermediate in the preparation of ponesimod of formula I or its pharmaceutically acceptable salts.

In accordance another embodiment, the present invention provides a process for the purification of ponesimod or its pharmaceutically acceptable salts which comprises of crystallizing ponesimod or its pharmaceutically acceptable salts from a suitable solvent or mixture thereof.

In accordance with further embodiment, the present invention provides pharmaceutical composition comprising ponesimod or its pharmaceutically acceptable salts prepared according to the process of the present invention and at least one pharmaceutically acceptable excipients.

Detailed description of the invention:

Accordingly, the present invention provides a process for the preparation of Ponesimod of formula I via novel intermediates.

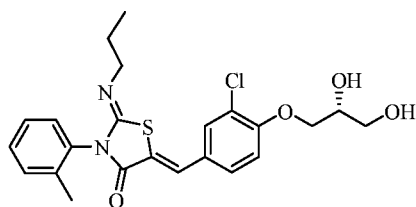
The term “suitable base/base” used herein the present invention until unless specified is selected from inorganic bases like “alkali metal hydroxides” such as lithium hydroxide, sodium hydroxide, potassium hydroxide and the like; “alkali metal carbonates” such

- sodium carbonate, potassium carbonate, lithium carbonate and the like; “alkali metal bicarbonates” such as sodium bicarbonate, potassium bicarbonate, lithium bicarbonate and the like; “alkali metal hydrides” such as sodium hydride, potassium hydride, lithium hydride and the like; “alkali metal alkoxides” such as sodium methoxide, sodium ethoxide, sodium tert-butoxide, potassium methoxide, potassium ethoxide, potassium tert-butoxide and the like, ammonia and organic bases such as triethylamine, methylamine, ethylamine, sodium acetate, potassium acetate, 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), 1,5-diazabicyclo(4.3.0)non-5-ene (DBN), lithiumdiisopropylamine (LDA), n-butyl lithium, tribenzylamine, isopropyl amine, diisopropylamine (DIPA), diisopropylethyl amine (DIPEA), N-methylmorpholine (NMP), N-ethylmorpholine, piperidine, dimethyl amino pyridine (DMAP), morpholine, pyridine, 2,6-lutidine, 2,4,6-collidine, imidazole, 1-methylimidazole, 1,2,4-triazole, 1,4-diazabicyclo [2.2.2]octane (DABCO) or mixtures thereof.
- 15 The term “suitable solvent” used herein the present invention until unless specified is selected from, but are not limited to “alcoholic solvents” such as methanol, ethanol, isopropyl alcohol, n-propanol, butanol and the like; “ester solvents” such as ethyl acetate, methyl acetate, n-butyl acetate, isobutyl acetate, sec-butyl acetate, isopropyl acetate and the like, “ether solvents” such as tetrahydrofuran, diethyl ether, methyl tert-butyl ether, dioxane and the like; “hydrocarbon solvents” such as toluene, xylene, cyclohexane, hexane, heptane, n-pentane, petroleum ether and the like; “chloro solvents” such as dichloromethane, ethylene dichloride, carbon tetrachloride, chloroform and the like; “polar aprotic solvents” such as dimethylformamide, dimethylacetamide, dimethylsulfoxide, N-methyl pyrrolidone and the like; “nitrile solvents” such as acetonitrile and the like; “ketone solvents” such as acetone, methyl isobutyl ketone, methyl ethylketone and the like; and water.

The term “Cu catalyst” used in the present invention until unless specified is selected from, but are not limited to CuSO_4 anhydrous, $\text{CuSO}_4 \cdot 3\text{H}_2\text{O}$, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{CuSO}_4 \cdot 7\text{H}_2\text{O}$, CuCl , CuBr , CuI , Cu_2O , $\text{Cu}(\text{OAc})_2$ and the like.

The term “suitable acid” used herein the present invention until unless specified is selected from, but are not limited to inorganic acids such as hydrochloric acid, hydrobromic acid, sulfuric acid, nitric acid, phosphoric acid etc; organic acids such as formic acid, acetic acid, maleic acid, malic acid, tartaric acid, oxalic acid, trifluoroacetic acid, methane sulfonic acid, p-toluene sulfonic acid and the like.

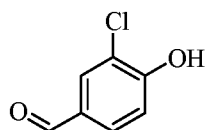
In a first embodiment, the present invention provides a process for the preparation of Ponesimod of formula I or its pharmaceutically acceptable salts,



Formula I

which comprises:

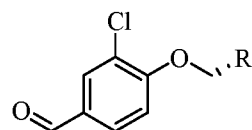
- a) reacting a compound of formula XII with a compound of general formula XIII in the presence of a base in a suitable solvent to provide a compound of general formula IV; wherein R is selected from oxirane or 2,2-dialkyl-1,3-dioxolane and X is halogen;



Formula XII

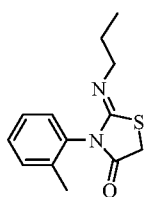


Formula XIII

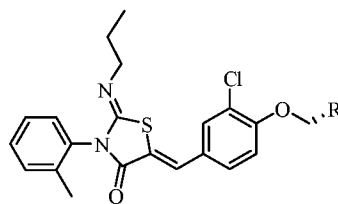


Formula IV

- b) condensing the compound of general formula IV with a compound of formula VI in the presence of a base in a suitable solvent to provide a compound of general formula II, wherein R is selected from oxirane or 2,2-dialkyl-1,3-dioxolane, and



Formula VI



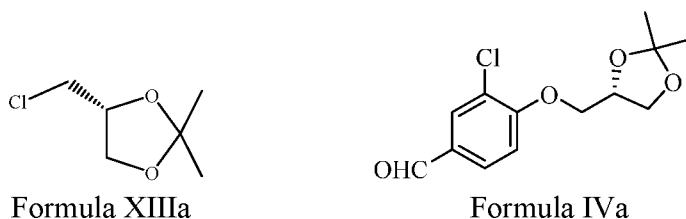
Formula II

- c) converting the compound of general formula II into Ponesimod of formula I or its pharmaceutically acceptable salts.

The starting compounds of formula XII & XIII are known in the art and commercially available; the same can be prepared by the known methods.

The step a) of aforementioned process involves the reaction of a compound of formula XII with a compound of general formula XIII, wherein R is selected from oxirane or 2,2-dialkyl-1,3-dioxolane; where alkyl is selected from methyl, ethyl or propyl and X is halogen selected from Cl, Br, I and F; in the presence of a base in a suitable solvent under appropriate reaction conditions to provide a compound of general formula IV, wherein R is selected from oxirane or 2,2-dialkyl-1,3-dioxolane. The said reaction may be carried out in the presence of an additional catalyst such as iodine source selected from iodine, potassium iodide and the like. The suitable base is selected from inorganic bases such as alkali metal hydroxides, alkali metal carbonates/bicarbonates, alkali metal alkoxides and alkali metal hydrides and/or mixtures thereof; and the suitable solvent is selected from alcohols, polar aprotic solvents and/or mixtures thereof. The step a) reaction is carried out at a suitable temperature of about 25°C to about reflux temperature of the solvent used for a sufficient period of time till completion of the reaction, preferably for about 15 to 20 hrs at reflux temperature.

In a preferred embodiment, the reaction of a compound of formula XII with a compound of formula XIII, wherein R is 2,2-dimethyl-1,3-dioxolane and X is Cl, represented by formula XIIIa, in the presence of a base such as potassium carbonate in a polar aprotic solvent such as dimethylformamide to provide a compound formula IV, wherein R is 2,2-dimethyl-1,3-dioxolane, represented by formula IVa.



The step b) of aforementioned process involves the condensation of compound of general formula IV with a compound of formula VI in the presence of a base selected from organic or inorganic bases, in a suitable solvent under appropriate reaction conditions to provide a compound of general formula II, wherein R is selected from oxirane or 2,2-dialkyl-1,3-dioxolane. The suitable base is selected from sodium acetate, potassium acetate,

The step a) of aforementioned process involves the condensation of compound of formula XV with a compound of formula V or compound of formula VI can be carried out in the presence of a base in a suitable solvent under appropriate reaction conditions to provide a corresponding compound of formula XVI or compound of formula XVII.

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The step b) of aforementioned process involves conversion of compound of formula XVI and compound of formula XVII into compound of formula IIA. The compound of formula XVI is first converted into compound of formula XVII by reacting with n-propylhalide in the presence of a base in a suitable solvent under appropriate condition; then compound of formula XVII subjected to chiral sharpless epoxidation under appropriate reaction conditions to provide compound of formula IIA.

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The step c) of aforementioned process involves the conversion of the compound of formula IIA into Ponesimod of formula I can be carried out by reacting with a base in a suitable solvent under appropriate reaction conditions. Optionally, Ponesimod of formula I is further converted into pharmaceutically acceptable salt by reaction with a suitable acid in a suitable solvent.

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In a further embodiment, the present invention provides a novel compounds of formula II, formula III, formula IVA (R is oxirane), formula VII, formula VIII, formula IX, formula X, formula XVI and formula XVII.

20

In a further embodiment, the present invention provides use of compounds of formula II, III, IVA (R is oxirane), VII, VIII, IX, X, XVI and XVII as an intermediate in the preparation of Ponesimod of formula I or its pharmaceutically acceptable salts.

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In a further embodiment, the present invention provides a process for the purification of Ponesimod or its pharmaceutically acceptable salts which comprises of crystallizing Ponesimod or its pharmaceutically acceptable salts from suitable solvent or mixture thereof.

30

In a further embodiment, the present invention provides pharmaceutical composition comprising Ponesimod prepared according to the process of the present invention and at least one pharmaceutically acceptable excipients.

Examples:

The process details of the invention are provided in the examples given below, which are provided by way of illustration only and therefore should not be construed to limit the scope of the invention.

5

Example-1: Preparation of (4*R*)-4-(chloromethyl)-2,2-dimethyl-1,3-dioxolane (XIIIa):

To clean and dry RBF, acetone (1 L), (2*R*)-3-chloropropane-1,2-diol (100 g), 2,2-dimethoxypropane (200 g) followed by CuSO₄.5H₂O (1.0 g) was added at 25-30°C and stirred for 2 hours. After completion of the reaction, acetone was distilled off completely under reduced pressure below 40°C. Ethyl acetate and DM water was added to the obtained residue at 25-30°C, stirred and both layers were separated. The aqueous layer extracted with ethyl acetate; the organic layers washed with water followed by sodium chloride solution, dried over sodium sulfate and distilled off under vacuum below 40°C to get the title compound. Yield: 95 g.

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Example-2: Preparation of 3-chloro-4-((4*R*)-2,2-dimethyl-[1,3]dioxolan-4-yl methoxy)-benzaldehyde (IVa):

To a clean and dry RBF, 3-chloro-4-hydroxybenzaldehyde (25 g), dimethyl formamide (250 ml), potassium carbonate (44 g), and (4*R*)-4-(chloromethyl)-2,2-dimethyl-1,3-dioxolane (72 g) was added at 25-30°C and stirred for 10 mins. The reaction mass was heated to 155-160°C and stirred for 18 hrs at the same temperature. After completion of reaction, the reaction mass was cooled to 25-30°C, filtered and washed with ethyl acetate. DM water and ethyl acetate was added to the filtrate, stirred and the layers were separated. The aqueous layer extracted with ethyl acetate; the organic layers combined, washed with water, followed by sodium chloride solution, dried over sodium sulphate and then distilled off completely under reduced pressure at 40-45°C to get the title compound. Yield: 66 g.

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Example-3: Preparation of *o*-tolyl isothiocyanate (XX):

To a solution of potassium carbonate (515 g) in water (2 L) at 0-5°C, *o*-toluidine (200 g) followed by carbondisulfide (275 g) were added drop wise at 0-5°C and stirred for 16 hrs at 25-30°C. Then, potassium carbonate (257 g) was added lot wise to the reaction mass followed by sodium persulfate solution (444 g in 1 L water) and was added dropwise at 0-

30

5°C. The reaction mass temperature was raised to 25-30°C and stirred for 2 hrs. After reaction completion, the reaction mass was filtered and pet ether (500 ml) was added to the filtrate and stirred. Both organic and aqueous layers were separated; the aqueous layer extracted with pet ether; the organic layers combined; washed with water followed by sodium chloride solution and dried over sodium sulphate then distilled off completely under reduced pressure to get the title compound. Yield: 195 g.

Example-4: Preparation of (Z)-2-(propylimino)-3-o-tolylthiazolidin-4-one (VI):

To a solution of o-tolylisothiocynate (10 g) in dichloromethane (100 ml), n-propylamine (5.23 g) was added dropwise at 20°C under nitrogen atmosphere and stirred for 30 mins. Then, chloroacetyl chloride (9.15 g) was added dropwise to the reaction mass at -5 to 0°C followed by triethylamine (14.2 g) and stirred for 30 mins at -5-0°C. After reaction completion, hydrochloric acid (300 ml) was added to reaction mass then stirred for 15 mins. Both organic and aqueous layers separated; aqueous layer washed with dichloromethane and the pH was adjusted to 8.0 with sodium bicarbonate solution. Then the reaction mass was extracted into dichloromethane and distilled off the solvent to get the title compound. Yield: 15.3 g.

Example-5: Preparation of n-propyl isothiocyanate (XXII):

To a solution of potassium carbonate (466 g) in water (1 L), n-propylamine (100 g) followed by carbon disulfide (256 g) was added drop wise to the reaction mass at 0-5°C; then the reaction mass temperature was raised to 25-30°C and stirred for 18 hours. Potassium carbonate (232 g) followed by solution of sodium persulfate (402 g) was drop wise added to the reaction mass at 0-5°C. Then the reaction mass temperature was raised to 25-30°C and stirred for 4 hours. After reaction completion, the reaction mass was filtered and petroleum ether (300 ml) was added to the filtrate. The organic and aqueous layers separated from the filtrate; aqueous layer extracted with Pet. Ether; the organic layers combined; washed with water followed by sodium chloride solution, dried over sodium sulphate and distilled off completely under reduced pressure at 40°C to get the title compound. Yield: 80 g.

Example-6: Preparation of (Z)-2-(propylimino)-3-o-tolylthiazolidin-4-one (VI):

To a solution of n-propyl isothiocyanate (10 g) in dichloromethane (100 ml), o-toluidine (12.5 g) was added dropwise at 20°C under nitrogen atmosphere and the stirred for 30 mins. The reaction mass was cooled to -5-0°C and chloroacetyl chloride (9.15 g) followed by pyridine (16.2 g) was added and stirred for 30 mins at -5-0°C. After the reaction completion, the reaction mass was quenched with hydrochloric acid and stirred for 10-15 mins. Organic and aqueous layers separated; aqueous layer washed with dichloromethane and pH was adjusted to 8.0 with sodium bicarbonate. Then the reaction mass extracted with dichloromethane and concentrated to get the title compound. Yield: 10 g.

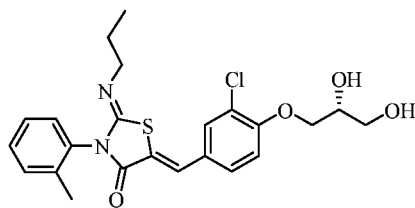
Example-7: Preparation of (2Z,5Z)-5-(4-(((R)-2,2-dimethyl-1,3-dioxolan-4-yl)methoxy)-3-chlorobenzylidene)-2-(propylimino)-3-o-tolylthiazolidin-4-one (IIa):

To a solution of o-tolylisothiocyanate (10 g) in dichloromethane (100 ml), n-propyl amine (5.23 g) was added dropwise at 20°C under nitrogen atmosphere and the maintained for 30 mins. Chloroacetyl chloride (9.15 g) was added dropwise at -5 to 0°C, followed by triethylamine (14.2 g) and stirred for 30 mins. After the reaction completion, the reaction mass was distilled off completely at 60°C. Isopropanol (110 ml), sodium acetate (7 g), 3-chloro-4-((4R)-2,2-dimethyl-[1,3]dioxolan-4-yl methoxy)-benzaldehyde (IVa) (20 g) was added to the residue at 60°C, stirred for 30 mins and further stirred for 6 hours at 80-85°C. After completion of the reaction, the reaction mass was cooled to 25-30°C, filtered, and the filtrate was distilled off completely. Ethanol (60 ml) was added to the obtained compound, heated to 70°C; DM water (60 ml) was added to it and stirred for 30 mins. Then the reaction mass was cooled to 25-30°C and stirred for 14 hrs. The obtained solid was filtered and dried to get the title compound. Yield: 12.5 g.

Example-8: Preparation of (2Z,5Z)-5-(4-(((R)-2,2-dimethyl-1,3-dioxolan-4-yl)methoxy)-3-chlorobenzylidene)-2-(propylimino)-3-o-tolylthiazolidin-4-one (IIa):

To a solution of n-propyl isothiocyanate (10 g) in dichloromethane (100 ml), o-toluidine (14 g) was added dropwise at 20°C under nitrogen atmosphere and the reaction mass was maintained for 30 mins. The reaction mass was cooled to -5 to 0°C and chloroacetyl chloride (13.3 g) was dropwise added to it at -5 to 0°C, followed by pyridine (16.2 g) and stirred for 30 mins. After completion of the reaction, the reaction mass was

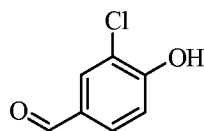
3. A process for the preparation of Ponesimod of formula I or its pharmaceutically acceptable salts,



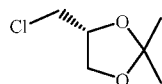
Formula I

5 which comprises:

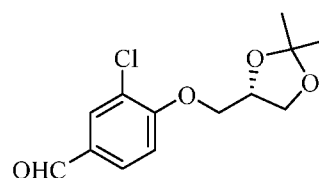
- a) reacting a compound of formula XII with a compound of formula XIIIa in the presence of a base in a solvent to provide a compound of formula IVa,



Formula XII

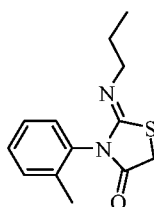


Formula XIIIa

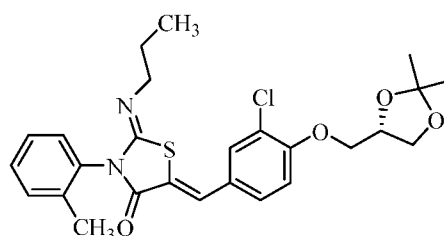


Formula IVa

- 10 b) condensing the compound of formula IVa with a compound of formula VI in the presence of a base in a solvent to provide a compound of formula IIa, and



Formula VI

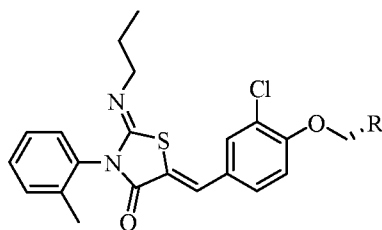


Formula IIa

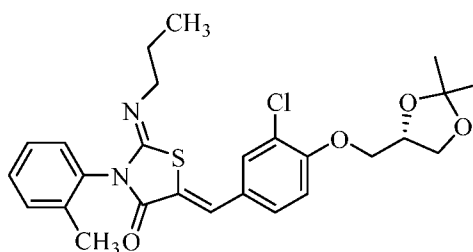
- 15 c) converting the compound of formula IIa into Ponesimod of formula I or its pharmaceutically acceptable salts.

4. The process as claimed in claim 1 and claim 3, wherein in step a) the suitable base is selected from inorganic bases such as alkali metal hydroxides, alkali metal carbonates/bicarbonates, alkali metal alkoxides and alkali metal hydrides and/or mixtures thereof; and the suitable solvent is selected from alcohols, polar aprotic solvents and/or mixtures thereof.
- 20

5. The process as claimed in claim 1 and claim 3, wherein in step b) the suitable base is selected from sodium acetate, potassium acetate, triethylamine, diisopropylamine (DIPA), diisopropylethyl amine (DIPEA), N-methylmorpholine (NMP), N-ethylmorpholine, piperidine, dimethyl amino pyridine (DMAP), morpholine, pyridine, 2,6-lutidine and/or mixtures thereof; and the solvent is selected from alcohols, hydrocarbon solvents, acetic acid and/or mixtures thereof.
6. The process as claimed in claim 1 and claim 3, wherein in step c) the conversion of general formula II or formula IIa into Ponesimod of formula I is carried out by treating compound of formula II or IIa with a suitable acid selected from hydrochloric acid (HCl), hydrobromic acid, sulfuric acid, nitric acid, phosphoric acid, acetic acid, maleic acid, malic acid, tartaric acid, oxalic acid, trifluoroacetic acid, methane sulfonic acid and/or p-toluene sulfonic acid.
7. The process as claimed in claim 1 and claim 3, wherein in step a) the base is potassium carbonate and the solvent is dimethyl formamide; and in step b) the base is sodium acetate and the solvent is isopropyl alcohol.
8. A compound of formulae selected from,



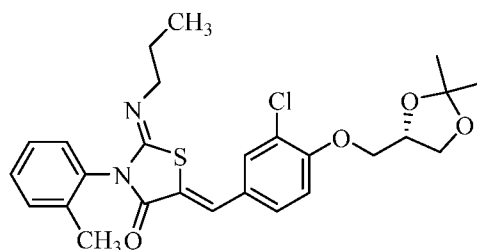
Formula II



Formula IIa

wherein R is selected from oxirane or 2,2-dialkyl-1,3-dioxolane.

9. A process for crystallization and purification of compound of formula IIa,



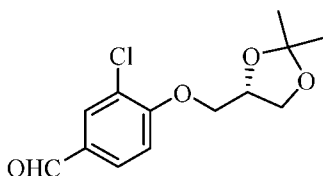
Formula IIa

which comprises

- 1) dissolving compound of formula IIa in a suitable solvent;
- 2) adding an suitable anti-solvent to step 1) solution; and
- 3) isolating pure compound of formula IIa.

10. The process as claimed in claim 9, wherein the suitable solvent in step 1) is alcohols selected from methanol, ethanol, propanol, butanol, isopropyl alcohol, t-butanol and/or mixtures thereof; and the suitable anti-solvent in step 2) is selected from water or hydrocarbons such as cyclohexane, hexane, heptane, n-pentane, petroleum ether and/or mixtures thereof.

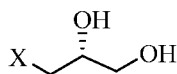
11. A process for the preparation of compound of formula IVa,



Formula IVa

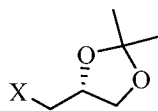
which comprises;

- a) reacting 1,2-diol compound of formula XVIII, wherein X is halogen or $-\text{OSO}_2\text{R}_1$; where R_1 is selected from alkyl group or aryl group with or without substitution;



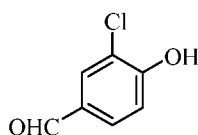
Formula XVIII

with 2,2-dimethoxypropane to provide a compound of formula XIIIb, wherein X is as defined above; and



Formula XIIIb

b) condensing the compound of formula XIIIb with a compound of formula XII,



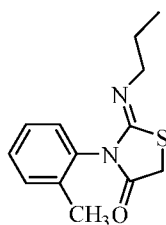
Formula XII

in the presence of a suitable base and a solvent to provide a compound of formula IVa.

12. The process as claimed in claim 11, wherein step a) reaction is carried out in the presence of a copper catalyst comprises anhydrous CuSO_4 , $\text{CuSO}_4 \cdot 3\text{H}_2\text{O}$, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{CuSO}_4 \cdot 7\text{H}_2\text{O}$, CuCl , CuBr , CuI , Cu_2O or $\text{Cu}(\text{OAc})_2$.

13. The process as claimed in claim 11, wherein step b) the suitable base is selected from inorganic bases such as alkali metal hydroxides, alkali metal carbonates/bicarbonates, alkali metal alkoxides and alkali metal hydrides and/or mixtures thereof; and the suitable solvent is selected from alcohols, polar aprotic solvents and/or mixtures thereof.

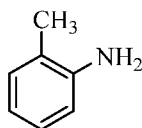
14. A process for the preparation of a compound of formula VI,



Formula VI

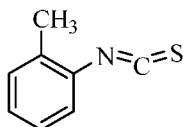
which comprises;

a) reacting a compound of formula XIX,



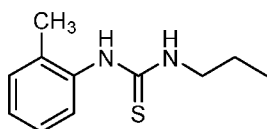
Formula XIX

with carbon disulfide in a suitable base to provide a *o*-tolylisothiocyanate of formula XX,



Formula XX

- 5 b) reacting the compound of formula XX with n-propylamine to provide a compound of formula XXI, and

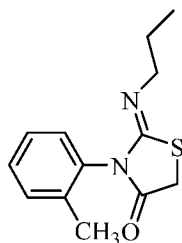


Formula XXI

- 10 c) cyclizing the compound of formula XXI obtained in step-(b) with 2-chloro-acetyl chloride in the presence of a suitable base and a solvent to provide a compound formula VI.

- 15 15. The process as claimed in claim 14, wherein the base in step a) is potassium carbonate; the base in step c) is triethylamine; and the solvent in step b) and c) is dichloromethane.

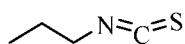
16. An alternate process for the preparation of a compound of formula VI,



Formula VI

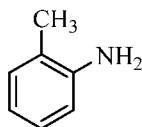
- 20 which comprises;

- a) reacting n-propylamine with carbon disulfide in a suitable base to provide a n-propyl isothiocyanate of formula XXII,



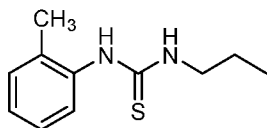
Formula XXII

- b) reacting the n-propyl isothiocyanate of formula XXII with a compound of formula XIX,



Formula XIX

- 5 to provide a compound of formula XXI, and



Formula XXI

- c) cyclizing the compound of formula XXI obtained in step-(b) with 2-chloro-acetyl chloride or 2-bromo-acetyl bromide in the presence of a suitable base and a solvent
10 to provide a compound formula VI.

17. The process as claimed in claim 16, wherein the base in step a) is potassium carbonate; the base in step c) is triethylamine or pyridine; and the solvent in step b) and c) is dichloromethane.

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18. A process for the preparation of Ponesimod of formula I, which comprises,

- 1) preparing the compound of formula VI or the compound of formula IVa according to the present invention process as described above; and
- 2) converting the compound of formula VI or the compound of formula IVa into
20 Ponesimod of formula I.